Journal of Sports Sciences

ISSN: (Print) (Online) Journal homepage: https://www.tandfonline.com/loi/rjsp20

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To cite this article: Andrew R. Novak, Franco M. Impellizzeri, Arjav Trivedi, Aaron J. Coutts & Alan McCall (2021): Analysis of the worst-case scenarios in an elite football team: Towards a better understanding and application, Journal of Sports Sciences, DOI: 10.1080/02640414.2021.1902138

To link to this article: https://doi.org/10.1080/02640414.2021.1902138

Published online: 10 Apr 2021.

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JOURNAL OF SPORTS SCIENCES
https://doi.org/10.1080/02640414.2021.1902138

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SPORTS PERFORMANCE

Analysis of the worst-case scenarios in an elite football team: Towards a better understanding and application

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ABSTRACT

This study investigated the variability in the worst-case scenario (WCS) and suggested a framework to improve the definition and guide further investigation. Optical tracking data from 26 male players across 38 matches were analysed to determine the WCS for total distance, high-speed running (>5.5 m.s^{-1}) and sprinting (>7.0 m.s^{-1}) using a 3-minute rolling window. Position, total output, previous epoch, match half, time of occurrence, classification of starter vs substitute, and minutes played were modelled as selected contextual factors hypothesized to have associations with the WCS. Linear mixed effects models were used to account for cross-sectional observations and repeated measures. Unexplained variance remained high (total distance R^2 = 0.53, high-speed running R^2 = 0.53 and sprinting R^2 = 0.40). Intra-individual variability was also high (total distance CV = 4.6–8.2%; high-speed CV = 15.6–37.8% and Sprinting CV = 21.1–76.4%). The WCS defined as the maximal physical load in a given time-window, produces unstable metrics lacking context, with high variability. Furthermore, training drills targetting this metric concurrently across players may not have representative designs and may underprepare athletes for complete match demands and multifaceted WCS scenarios. Using WCS as benchmarks (reproducing similar physical activity for training purposes) is conceptually questionable.

ARTICLE HISTORY
Accepted 9 March 2021

KEYWORDS
GPS; team sports; soccer; physical performance; player monitoring

Introduction

The physical characteristics of team sports athletes have been of interest to researchers and practitioners for several decades (Messersmith & Corey, 1931; Reilly, 1976). By assessing athlete match activities to identify the physical and physiological determinants of performance, it is possible to define training targets to design training sessions (Impellizzeri et al., 2019). Historically, physiological measures have been used to understand these characteristics (Bangsbo, 1994). More recently, technological innovations have enabled relatively accurate quantification of physical activity, from which physiological requirements can be estimated e.g., Global Navigations Satellite Systems and Optical Tracking (Low et al., 2019; Weaving et al., 2019). These systems capture players' positional coordinates, and variables such as distance, velocity, and acceleration are calculated. However, to understand the physiological demands imposed by the external training load, internal training load measures are also necessary. When interpreted together, the external and internal training loads can be used to understand the physiological requirements of physical performance, and also monitor whether a favourable (or unfavourable) internal response occurred.

More recently, external load measures from matches have been used to inform the design of specific training exercises such as small-sided games that closely replicate the activities of matches (Delaney, Duthie et al., 2016; Delaney, Thornton et al., 2016; Furlan et al., 2015; Johnston et al., 2019; Weaving et al., 2019). The assumption being that by reproducing the external load of periods throughout a game, such training provides a sufficient stimulus to induce physiological outcomes that underpin the physical performance. Initially, analysis of match-based external loads involved capturing the total activities of match play i.e. total distance or total high-speed running, and often drew comparisons between match periods, positional groups, competition levels, or relative time spent at various intensities (Deutsch et al., 1998; Duthie et al., 2005; McLellan et al., 2011; Sirotic et al., 2009). However, it was later suggested that the average match activities have limited relevance in athlete preparation, and that applying such values towards training may not expose players to the most intense periods which occur intermittently throughout matches (Delaney et al., 2015). It has not yet been clarified whether this approach is to understand the demands, and hence the physiological systems to train, or to develop training exercises (or both). Some studies suggested these intense periods can be used to inform generic training prescription (Delaney, Thornton et al., 2016) while others suggested these intense periods as benckmarks for exercise replication (Delaney et al., 2015; Oliva-Lozano et al., 2021). From a recent Delphi survey, the latter seems to be a common interpretation of the use of the most demanding phases in professional football (McCall et al., 2020)

Several methods have since been proposed to capture such periods – often referred to by terms such as the worst-case

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scenario (Cunningham et al., 2018; Fereday et al., 2020; Gabbett et al., 2016; Johnston et al., 2014; Reardon et al., 2017), peak match demands (Whitehead et al., 2018), peak locomotor demands (Weaving et al., 2019), peak movement and collision demands (Johnston et al., 2019) and duration-specific running demands (Delaney, Duthie et al., 2016; Delaney, Thornton et al., 2016). While there are numerous terms and methods used, the concepts and intended practical applications are similar and can be discussed together, hereafter referred to as the worst-case scenario (WCS).

Methods to capture the WCS include: analysing periods of repeated high-intensity efforts (Johnston et al., 2014), the longest periods of ball-in-play (Cunningham et al., 2018; Reardon et al., 2017), analysing shorter periods of fixed durations (Cunningham et al., 2018), or using a moving average analysis technique across many time periods (Cunningham et al., 2018; Delaney, Duthie et al., 2016; Delaney, Thornton et al., 2016; Furlan et al., 2015; Johnston et al., 2019; Varley et al., 2012). This moving average analysis has been applied to numerous external load variables (e.g., distance, average speed, high-speed running, average acceleration, metabolic power, and collisions) and across numerous durations (e.g., typically 10 s-10 min) (Delaney, Thornton et al., 2016; Weaving et al., 2019), (Delaney, Duthie et al., 2016), (Delaney et al., 2015)' (Cunningham et al., 2018), (Johnston et al., 2014). In all cases, the intent appears the same – to capture the most intense physical activities experienced during match-play, so that the same stimulus can be reproposed to players in order to elicit positive outcomes (McCall et al., 2020). Typically, the WCS has been compared between positional groups and between match halves to show that the WCS is influenced by these factors (Delaney, Duthie et al., 2016; Delaney, Thornton et al., 2016; Weaving et al., 2019). We believe that the WCS may be influenced by numerous additional contextual factors e.g., the actual time of occurrence within the match half, whether the player was a starter or substitute player, the amount of minutes played by the athlete, and the amount of activity completed in the period immediately preceding the WCS. To clarify, a player entering the field at the end of a match and with low prior activity may achieve a higher peak WCS value than the same player who expects to play the entire match and complete relatively high activity throughout. These contextual factors should be considered together and we believe that the concept of WCS has not yet been adequately defined, and an appropriate framework has not been proposed to test the existence and usefulness of the WCS, other than generic explanations. Considering the suggestion to use (and the actual use in practice of) the WCS as benchmarks, the examination of its suitability as a refence quantity to be replicated in training needs conceptual and critical exploration.

The overall aim of this study was to investigate the WCS in football as defined by a rolling time windows. Though prior studies have reported positional and periodic differences (differences between match halves) for various univariate metrics in Rugby League (Delaney, Duthie et al., 2016; Weaving et al., 2019) and Rugby Union (Delaney, Thornton et al., 2016), and that practitioners are using the method to monitor high-speed running and sprinting metrics (McCall et al., 2020), we aimed to quantify the variability of WCS when such analyses are implemented. We hypothesized that the WCS, as defined through rolling time windows, is highly variable, influenced by numerous contextual factors, and, consequently, of limited value to practioners when designing training programmes. A second aim of the study was to propose a potential framework to better define and investigate the WCS.

Methods

Study design and subjects

A retrospective, cross-sectional study was conducted using data from 26 male professional football (soccer) players (age: 28 ± 4 y; height: 182 ± 6 cm; body mass: 78.8 ± 6.2 kg). These players competed for an English Premier League team across 38 matches of the 2018/19 season.

Data collection, calculation & processing

Athlete positional coordinates were collected from each match, sampled at 25 FPS using the TRACAB Gen4 optical tracking system (ChyronHego, New York, USA) which has been validated previously (c.f. Linke et al. 2015) (Linke et al., 2020). We calculated athlete speeds from the raw positional data, and outliers of unphysical human speeds were replaced with the mean of adjacent data. For each player, for each match, within each period, the WCS was defined as the peak 3-min total (WCS_{TD}), high-speed running (≥ 5.5 m/s; WCS_{HSR}) and sprinting (≥ 7.0 m/s; WCS_{SP}) distances, calculated using rolling windows (we also analysed a 5-min rolling window; however, after obtaining similar results we omitted analysis and reporting of them for brevity). One observation was collected for each player, per match half, resulting in 33 ± 22 (range: 1–66) observations per player. After the WCS was identified, the preceding 3-min period was calculated as the pre-epoch. Contextual variables were collected including positional group, classification of starter vs. substitute, match half ($1^{st}/2^{nd}$), uncoupled total activity for the match-half (to avoid spurious relationships, WCS for the match-half was subtracted from the total activity for the match-half), minutes since the beginning of the match half, total minutes played per match-half, and the pre-epoch.

Players were removed if they were goalkeepers, or participated in <5 matches, leaving 21 players for the final analysis. Players' primary positional groups consisted of Central Midfielders (CM; n = 7), Central Backs (CB; n = 5), Wide Midfielders (WM; n = 2), Fullbacks (FB; n = 4), and Strikers (ST; n = 3). While approval for the study was obtained from the reference club involved, these data arose as a condition of employment in which player data were routinely collected over the course of the competitive season. Therefore, usual appropriate ethics committee clearance was not required. (Winter & Maughan, 2009). To ensure team and player confidentiality, all data were anonymized before analysis.

Statistical analysis

All statistical analyses were conducted using R (R Core Team. R, 2019). Presence of outliers was investigated for values falling outside of 2.2 multiplied by the interquartile range. We

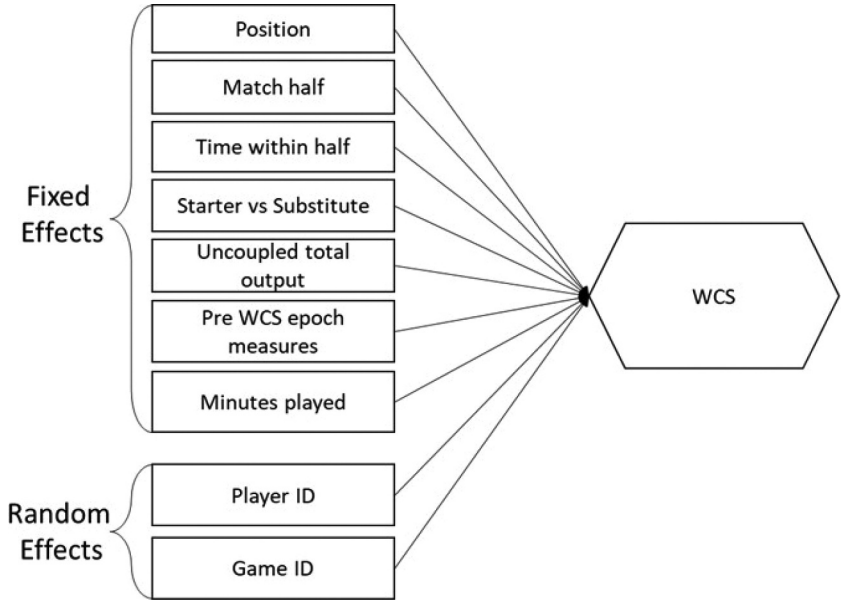

Figure 1. Our model design. Fixed effects were hypothesized to have associations with the Worst Case Scenario. Random effects indicate sources of variance attributed to repeated measures obtained from the sample.

Table 1. Distance covered per positional group for total, high-speed running and sprinting activities three-minute Worst Case Scenario.

	Records (n)	Total distance (m) Mean (SD) – CV%	High speed running (m) Mean (SD) – CV%	Sprinting (m) Mean (SD) – CV%
Central Midfielders	292	500.0 ± 33.0– 6.6%	91.1 ± 26.2– 28.8%	32.8 ± 18.4– 56.1%
Centre Backs	178	439.1 ± 29.3– 6.7%	66.4 ± 20.3– 30.6%	29.6 ± 15.1– 51.0%
Fullbacks	150	478.6 ± 30.9– 6.5%	102.5 ± 23.0– 22.4%	45.1 ± 15.8– 35.0%
Strikers	118	443.5 ± 29.2– 6.6%	97.4 ± 20.5– 21.0%	44.1 ± 18.4– 41.7%
Wide midfielders	94	474.8 ± 32.6– 6.9%	107.2 ± 25.3– 23.6%	46.6 ± 19.0– 40.8%

Key: CV = Coefficient of Variation; SD = Standard Deviation

analysed the WCS using a mixed effects model approach (Figure 1); one model for each dependent variable (WCS_{TD}, WCS_{HSR}, WCS_{SP}). Random effects were specified to account for the cross-sectional nature of player and match observations, as well as the repeated measures for individual players across matches. As all players did not compete in all matches, the player ID and match ID were modelled as crossed effects. A player's positional group was the position they played in the match i.e. not necessarily their nominal position. Continuous variables were standardized to facilitate comparisons across variables of differing scales.

All independent variables were included within all models. Multicollinearity was investigated for each model prior to analysis, and where two variables were correlated above 0.8, the variable with a lower correlation with the WCS was removed from the model. Positional group was a nominal fixed effect. Therefore, interpretations of positional group are provided relative to the reference value of Central Midfielders (the group with the most observations). We calculated coefficient estimates and 95% confidence intervals (CI) for each independent variable in each WCS model. To assess model fit, the models were each compared against a null model (with only random effects), using the Akaike Information Criterion (AIC). Model residuals were inspected for deviation from normality, and Cook's distance was calculated to investigate influential points (cut-off = 1). We calculated marginal and conditional R^2 values to assess the explained variance excluding and including random effects.

Results

Each model used 832 WCS observations. We detected multicollinearity between minutes played and uncoupled total output (r = 0.93 [95%CI: 0.92–0.94) for WCS_{TD}. Uncoupled total distance was retained while minutes played were removed from this model (c.f. Methods – Statistical Analysis) as it had a stronger correlation with WCS_{TD} (r = 0.25 [95% CI: 0.19– 0.31] vs r = −0.01 [95%CI: −0.08– 0.06). Average values and variability per positional group are presented in Table 1. Intra-individual variability is displayed in Figure 2. The variability in timing/occurrence of each WCS is presented in Figure 3 after filtering to only include observations in which players started on-field and completed the entire half.

WCS total distance, WCS_{TD}

The average intra-individual CV for total distance was 6.2% (range: 4.6– 8.2%). The specified model was an improvement over the null model (AIC = 1688.5 vs. 1864.6; $p < 0.001$). Marginal R^2 was 0.28 and conditional R^2 was 0.53, resulting

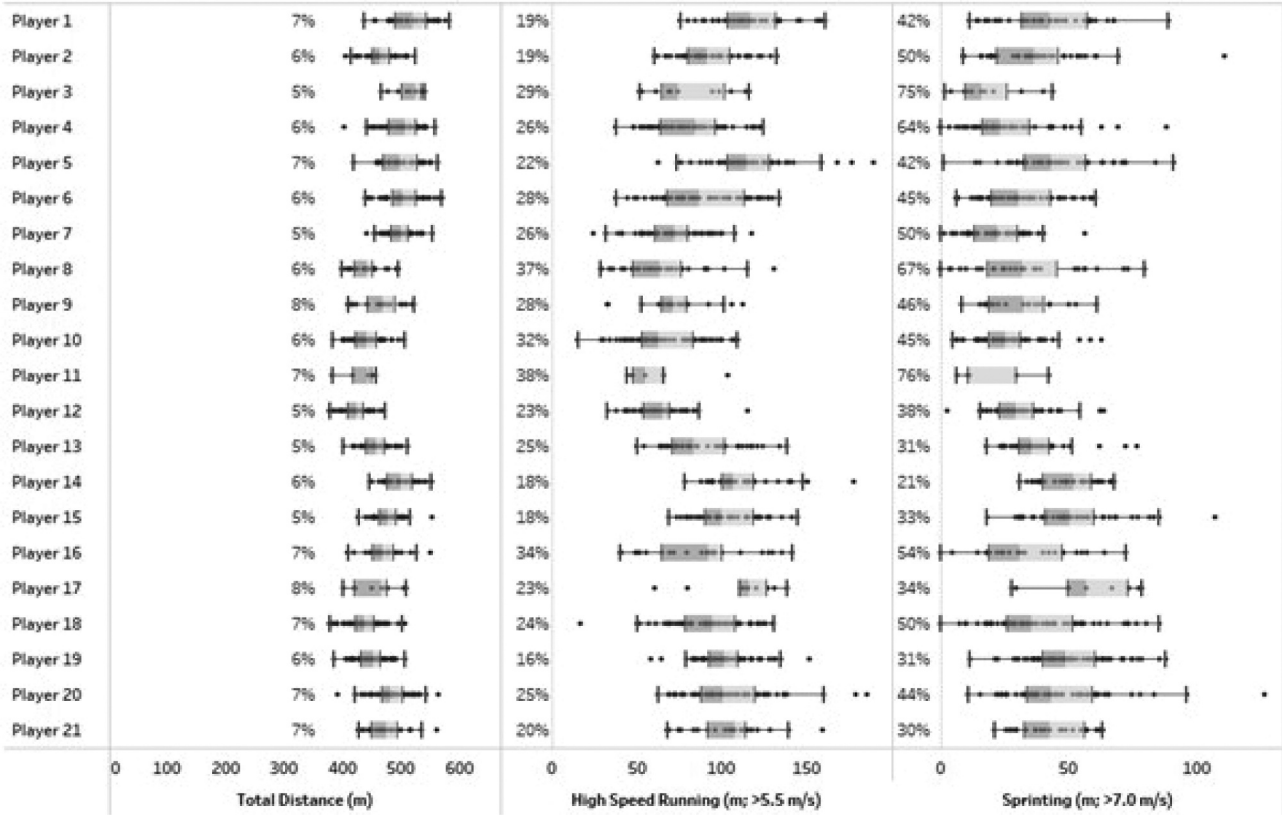

Figure 2. Individual variability (coefficient of variation as a percentage) of the Worst Case Scenario over one season across three measures using a three-minute rolling average window.

from variance across individual players (variance = 0.161; SD = 0.401) and matches (variance = 0.041; SD = 0.203). The effect of positional group was strong (Table 3), with CB, FB, ST and WM having lower WCS_{TD} values than CM (standardized β = −0.910 – −0.309, p = <0.001– 0.030). However, confidence intervals were wide indicating high variability within positional groups. Substitutes and players who completed higher uncoupled total loads tended to produce higher values of WCS_{TD}, while lower values of WCS_{TD} occurred later in the match (within the second half and/or later within the half). There was no effect of pre-epoch on WCS_{TD} (95% CI = −0.049–0.046; p = 0.940).

WCS high-speed running (> 5.5 m/s), WCS_{HSR}

The average intra-individual CV for WCS_{HSR} was 25.2% (range: 15.6– 37.8%). The specified model improved the AIC from 2008.3 to 1763.5 ($p < 0.001$). Marginal R^2 was 0.47 and conditional R^2 was 0.53, indicating some variance across individual players (variance = 0.033; SD = 0.181) and matches (variance = 0.025; SD = 0.158). There was less effect of positional group than for the WCS_{TD}, with only CB having lower WCS_{HSR} than CM (standardized β = −0.383 [95%CI: −0.604 – −0.161]; p = 0.003) (Table 4). A greater uncoupled total output was associated with higher WCS_{HSR} values (standardized β = 0.610 [95%CI: 0.532– 0.707]; $p < 0.001$) while a lower amount of minutes played was associated with higher WCS_{HSR} values for high-speed running (standardized β = −0.191 [95% CI: −0.282 – −0.117]; $p < 0.001$).

WCS sprinting (> 7.0 m/s), WCS_{SP}

The average intra-individual CV for sprinting was 46.1% (range: 21.1– 76.4%). The specified model improved the AIC from 2174.8 to 1941.7 ($p < 0.001$). Marginal R^2 was 0.37 and conditional R^2 was 0.40. There was some variance across individual players (variance = 0.029; SD = 0.171) and little variance across matches (variance = 0.004; SD = 0.060) (Table 5). There were no significant inter-positional differences, potentially due to the high intra-positional variance (37–55%; Table 1). However, higher WCS_{SP} values were associated with greater uncoupled total output (standardized β = 0.578 [95%CI: 0.509– 0.666; $p < 0.001$) and later occurrence in the half (standardized β = 0.059 [95%CI: 0.004– 0.116]; p = 0.038).

Discussion

This study investigated variability in the WCS using a common rolling window technique. Comparisons were made between positions as per previous methods, while also controlling for numerous additional contextual factors hypothesized to influence the WCS. Positional differences were identified as per previous research e.g., between central midfielders and all other positions for total distance (CM = 500.0 (SD: 33.0) m;

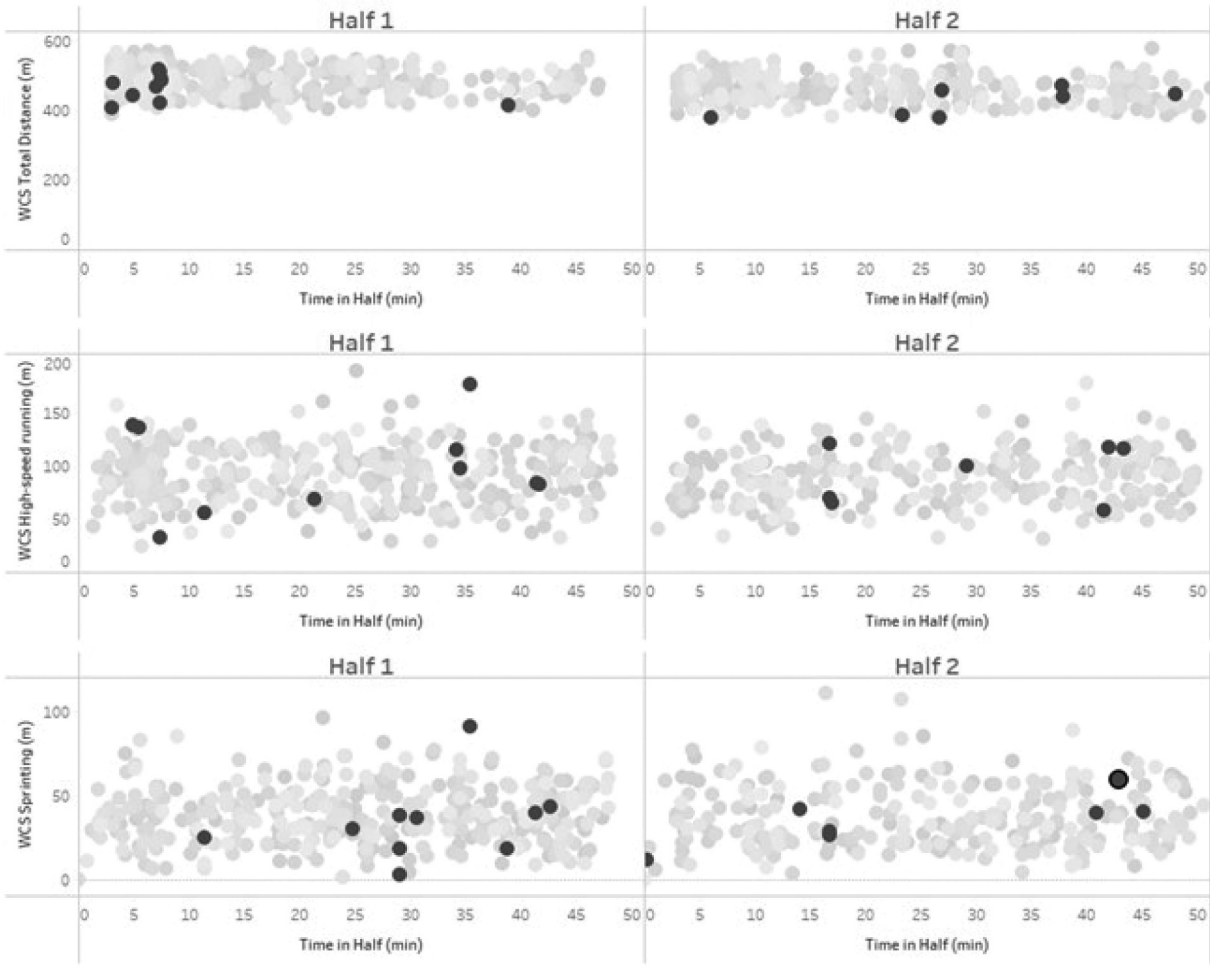

Figure 3. Variability in timing/occurrence of the Worst Case Scenario, with one match highlighted to visualize that Worst Case Scenario measured by different variables is not necessarily concurrent (only players who started and completed the entire match half are included).

CB = 439.1 (SD: 29.3) m; FB = 478.6 (SD: 30.9) m; ST = 443.5 (SD: 29.2); WM = 474.8 (SD: 32.6)). However, only CB were lower than CM for HSR (CM = 91.1 (SD: 26.2) m; CB = 66.4 (SD: 20.3) m) and no differences between CM and any other positional group were identified for sprinting.

For total distance, larger WCS_{TD} was associated with earlier occurrence in the match and half, higher uncoupled total output, and for substitute players. Comparatively, larger WCS_{HSR} was associated with fewer minutes played and higher uncoupled total output, while higher values of WCS_{SP} were associated with higher uncoupled total outputs and later occurrence within the half. These variables explained some variance within the WCS, yet unexplained variance remained high e.g., 0.47 (WCS_{TD} and WCS_{HSR}) to 0.60 (WCS_{SP}), and intra-positional variability ranging from 6.5% (WCS_{TD} for FB) to 56.1% (WCS_{SP} for CM). This is not surprising given that the intra-individual variability (CV) was high (4.6–8.2% for WCS_{TD}, 15.6–-37.8% for WCS_{HSR} and 21.1–76.4% for WCS_{SP}). This ultimately reflects metrics that are unstable and would be difficult to utilize in practice. This finding is pertinent to share with the football community, as the WCS represents one strategy of teams in an attempt to prepare players optimally for match-play by replicating the WCS, and the variability showed in this study raises questions about the use of WCS as benchmarks (McCall et al., 2020)

Interestingly, WCS_{TD} occurred earlier within the half while WCS_{SP} occurred later. Data plotted for individual games suggests that players may not achieve WCS for different metrics concurrently (Figure 3). Consequently, in training drills, the WCS may not be achieved for multiple players across multiple metrics within a single drill, unless the drill has been specifically designed to do so. While such drills might be designed with conditioning in mind (although such training studies have not been reported to our knowledge), further investigation is required to determine their utility from a skill acquisition perspective, i.e. if the WCS for multiple metrics and multiple players is achieved within a single drill, then the representative design may be questionable.

An important limitation we identified was that several players were only observed <5 times across the season. Given that some practitioners use 100% of the individual's WCS as a form of benchmark in training (McCall et al., 2020), it is difficult to determine what should be done with these players from a practical perspective. If a coach/scientist has only a handful of observations, our findings suggest they should not be confident that the values reflect any true physiological relevance for the player.

Table 2. The median and interquartile range of Worst Case Scenario (WCS), total outputs, WCS as a percentage of the total output, and the highest WCS for individual players for total distance, high-speed running and sprinting.

		Total Distance				High-speed running				Sprinting			
Player ID	n	WCS (m)	Total Output (m)	WCS (%)	Highest WCS (m)	WCS (m)	Total Output (m)	WCS (%)	Highest WCS (m)	WCS (m)	Total Output (m)	WCS (%)	Highest WCS (m)
Player 1	41	520 (53)	4305 (3282)	11 (9)	582	117 (29)	516 (448)	23 (13)	161	43 (26)	111 (102)	39 (17)	89
Player 2	43	463 (30)	5307 (1388)	9 (2)	523	91 (25)	469 (213)	22 (7)	132	36 (24)	97 (71)	36 (21)	111
Player 3	11	525 (33)	5830 (2441)	9 (6)	540	74 (37)	309 (170)	27 (10)	116	16 (16)	27 (28)	64 (19)	44
Player 4	57	506 (47)	5960 (456)	8 (1)	558	84 (32)	343 (160)	23 (10)	124	23 (18)	50 (54)	52 (32)	88
Player 5	43	497 (58)	5786 (1467)	9 (3)	562	114 (25)	606 (237)	19 (12)	189	43 (24)	124 (80)	37 (21)	91
Player 6	54	501 (39)	5990 (543)	9 (1)	570	87 (45)	407 (196)	20 (10)	134	30 (23)	67 (59)	43 (26)	60
Player 7	57	499 (28)	5869 (354)	9 (1)	553	72 (20)	332 (141)	22 (8)	118	22 (16)	43 (37)	49 (37)	56
Player 8	19	468 (48)	5463 (351)	9 (1)	521	71 (15)	329 (82)	21 (6)	112	32 (22)	66 (46)	45 (25)	61
Player 9	60	440 (34)	5230 (370)	8 (1)	505	63 (30)	304 (99)	22 (9)	109	25 (11)	57 (46)	43 (25)	63
Player 10	5	444 (31)	3639 (3360)	12 (17)	456	55 (18)	213 (188)	47 (30)	103	11 (19)	48 (34)	44 (61)	42
Player 11	49	423 (27)	5074 (378)	8 (0)	471	65 (15)	266 (101)	23 (5)	115	29 (13)	68 (31)	46 (19)	64
Player 12	31	442 (32)	5227 (543)	8 (1)	495	64 (28)	262 (201)	24 (8)	131	32 (32)	55 (69)	56 (30)	80
Player 13	42	457 (31)	5378 (391)	9 (0)	511	83 (30)	427 (145)	19 (4)	138	36 (12)	101 (53)	35 (14)	77
Player 14	35	500 (42)	5853 (325)	8 (0)	552	106 (19)	612 (172)	18 (5)	177	52 (19)	163 (98)	34 (11)	68
Player 15	45	477 (29)	5684 (447)	8 (1)	553	100 (29)	570 (131)	18 (3)	144	50 (19)	178 (104)	29 (14)	107
Player 16	26	467 (32)	5593 (2591)	9 (6)	550	92 (33)	443 (304)	20 (11)	141	31 (26)	71 (86)	35 (22)	72
Player 17	9	466 (56)	2005 (473)	21 (6)	508	116 (16)	299 (105)	37 (5)	138	57 (24)	120 (69)	49 (7)	79
Player 18	61	437 (31)	5046 (602)	9 (1)	506	94 (29)	480 (205)	21 (7)	130	36 (25)	101 (76)	38 (18)	85
Player 19	66	451 (33)	5058 (749)	9 (1)	506	101 (17)	551 (192)	18 (6)	151	49 (20)	178 (74)	27 (10)	88
Player 20	53	482 (37)	5326 (2441)	9 (6)	564	101 (32)	507 (232)	21 (10)	185	43 (25)	119 (111)	40 (18)	127
Player 21	25	473 (44)	5386 (1790)	9 (3)	561	108 (22)	592 (215)	18 (9)	159	43 (23)	149 (87)	27 (19)	63

Key: WCS = Worst Case Scenario

Table 3. Model for the Worst Case Scneario for 3-minute total distance.

Fixed Effects	Standardized estimate	95% confidence interval	T statistic	P value
(Intercept)	0.552	0.297– 0.801	4.382	<0.001
Position group – Centre Backs	−0.910	−1.256 – −0.573	−5.629	<0.001
Position group – Full Backs	−0.309	−0.594 – −0.031	−2.19	0.030
Position group – Strikers	−0.605	−0.961 – −0.272	−3.639	<0.001
Position group – Wide Midfielders	−0.406	−0.621 – −0.201	−3.816	<0.001
Starter vs substitute	0.650	0.446– 0.859	6.167	<0.001
Match half	−0.343	−0.438 – −0.248	−7.059	<0.001
Uncoupled total output	0.225	0.166– 0.285	7.417	<0.001
Time in half	−0.090	−0.142 – −0.037	−3.325	<0.001
Previous epoch	0.002	−0.049– 0.046	−0.076	0.940

Use of WCS in practice: challenges

There are several risks and limitations that should be considered using WCS data to inform training prescription. While specific training recommendations have not been reported (to our knowledge), we assume that the intention of reporting positional group comparisons has been to guide the development of training targets. However, as our data shows, there is high variability and WCS occurs in different phases of the matches, indicating the real demands can vary depending on when these scenarios occur. Additionally, whether this "benchmark" is an adequate stimulus

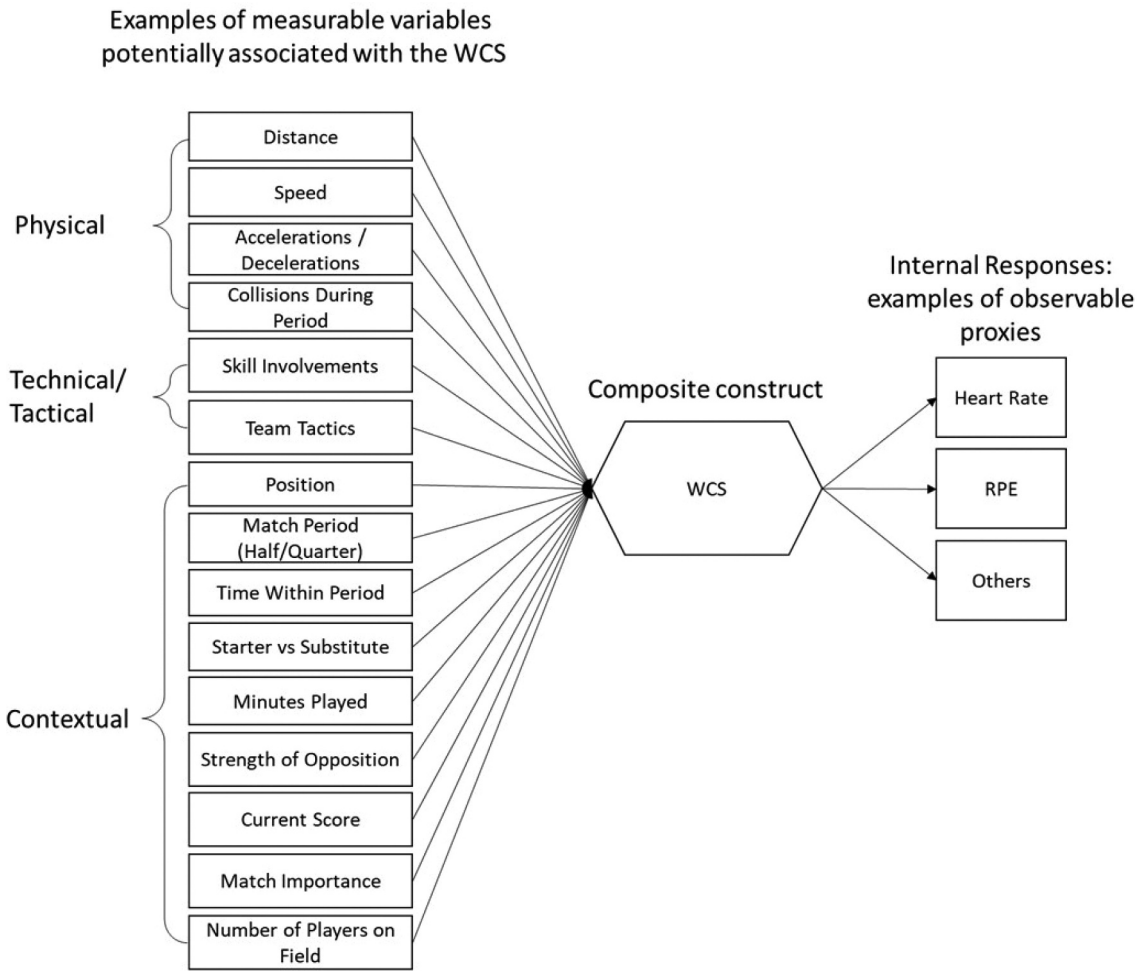

Figure 4. A framework to define and investigate the WCS as a composite construct induced by various combinations of contextual factors and external loads, and observable by measures of internal response.

to induce relevant physiological adaptations is questionable. For example, the values for WCS are relatively low e.g., across 3 min, the mean WCS_{TD} for CM was 500 m (2.77 m/s), while our threshold for high-speed running was twice this speed at 5.5 m/s. Mean WCS_{SP} was only 32.8 m across a 3 min period of play. This intensity seems unlikely "optimal" for inducing changes, or at least it is questionable. Similarly, 100 or 150 m at high-speed running is not necessarily a sufficient stimulus (e.g., much more may be needed). As a consequence, the use of such benchmarks in practice may provide limited improvements in physical characteristics. If players are only prepared to meet the typical WCS of their positional group or to meet their previous peak value, it would not be logical for them to perform successfully under atypical conditions. Targeting the WCS may present a recursive cycle, whereby players perform at the WCS, are then trained to target their prior WCS (or positional group average WCS) and may subsequently be unprepared for atypical conditions. By extension, the same could be said of many time-motion analyses from matches and practitioners should consider the conceptual limitations associated with this approach in planning training activities i.e. WCS, as also other quantitative match analysis measures, may help to understand relevant physical characteristics to train but their use as benchmarks is questionable and potentially misleading.

Furthermore, even if the WCS was a stable/reliable metric, it still only reflects external load for single (typically locomotor) variables (although investigation with respect to concurrent collisions has been reported) (Johnston et al., 2019), with multiple univariate interpretations required. This is problematic given that a true WCS (or even a more conservative definition of peak locomotor demands) is likely a far more complex, multivariate construct. For example, a 3-minute scenario in which a player achieves 470 m of total distance would be selected (and subsequently focused in training) by a univariate rolling analysis, in favour of a scenario in which a player achieves 460 m of total distance and 50 m of sprinting, after 80 min of match play and immediately after a high uncoupled total load. Univariate WCS targets without additional contextual information would likely underprepare players for multivariate scenarios. Furthermore, there could be numerous combinations of factors which produce a true WCS (i.e. an extreme internal response). Therefore, we provide a framework to assist in

Table 4. Model for the Worst Case Scneario for 3-minute high-speed running.

Fixed Effects	Standardized estimate	95% confidence interval	T statistic	P value
(Intercept)	0.010	−0.145– 0.163	0.122	0.904
Position group – Centre Backs	−0.383	−0.604 – −0.161	−3.236	0.003
Position group – Full Backs	0.101	−0.107– 0.304	0.922	0.362
Position group – Strikers	−0.036	−0.279– 0.210	−0.276	0.785
Position group – Wide Midfielders	0.114	−0.079– 0.327	1.101	0.272
Starter vs substitute	0.115	−0.101– 0.330	1.039	0.299
Match half	0.093	−0.003– 0.194	1.844	0.065
Uncoupled total output	0.610	0.532– 0.707	14.751	<0.001
Time in half	0.042	−0.008– 0.092	1.637	0.102
Previous epoch	0.028	−0.021– 0.079	1.112	0.266
Minutes played	−0.191	−0.282 – −0.117	−4.820	<0.001

Table 5. Model for the Worst Case Scneario for 3-minute sprinting.

Fixed Effects	Standardized estimate	95% confidence interval	T statistic	P value
(Intercept)	−0.043	−0.195– 0.103	−0.531	0.602
Position Group – Centre Backs	−0.044	−0.254– 0.167	−0.377	0.711
Position Group – Full Backs	0.034	−0.174– 0.255	0.295	0.77
Position Group – Strikers	0.026	−0.225– 0.288	0.186	0.854
Position Group – Wide Midfielders	0.121	−0.082– 0.363	1.078	0.282
Starter vs substitute	−0.126	−0.369– 0.119	−1.005	0.315
Match half	0.087	−0.021– 0.199	1.562	0.119
Uncoupled total output	0.578	0.509– 0.666	15.234	<0.001
Time in half	0.059	0.004– 0.116	2.080	0.038
Previous epoch	0.015	−0.041– 0.073	0.530	0.596
Minutes played	−0.066	−0.149– 0.004	−1.721	0.086

understanding and expanding on the WCS concept. Specifically, we propose that:

(1) The WCS appears to be highly individual. Therefore, we doubt that a true WCS could be determined without understanding the individual internal response i.e. two players completing the same scenario above would likely not respond the same and would not elicit the same physiological responses. Measures of internal load are required to capture this information and monitor training appropriately.
(2) A true WCS is likely to occur under multivariate conditions e.g., a relatively high volume of sprinting + high volume of acceleration/deceleration, collisions and/or skill involvements, later in the match for starting players, with high uncoupled total load. Also, numerous combinations of these conditions could induce a WCS i.e. an extreme internal response to various external and contextual factors. Preparing for one univariate (or even multivariate WCS) may not provide adequate preparation for another, yet understanding how physical and contextual factors combine in different ways to produce WCS responses may guide the development of robust training programmes.
(3) Training studies must be conducted to develop any confidence that training to the WCS is a suitable approach to improve player performance.

Considering these points, we hypothesize that various combinations of external load metrics and contextual factors combine to produce the WCS, and the intra-individual WCS could be observed via measures of internal response to multivariate conditions. We provide Figure 4 to help illustrate how the concept of WCS is multifactorial thus explaining some reasons for its variability. This visual format is consistent with structural equation modelling, which is just one statistical approach that can be used in modelling the WCS as a composite construct (See Kline, 2015 for an introduction to structural equation modelling). Furthermore, manifestations of the WCS could be observed also via monitoring appropriate internal responses. The observable internal response to WCS should be selected appropriate to the specific activity and setting, e.g., heart rate and RPE may be appropriate proxies for on-field team-sports activities. While measures such as heart rate are feasible in several contexts, other measures, such as RPE or lactate may be challenging to collect during real matches, but simulations may be an option providing a greater level of control to test the internal response to WCS as done in seminal studies (Bangsbo, 1994). Simulations are clearly relevant just to explore physiological responses to WCSs to understand and define training targets, once typical WCS has been identified.

Conceptual considerations

Defining high-intensity phases as WCS (or peak match demands) is misleading and whether this constitutes the actual WCS is questionable. The WCS is unlikely to be the peak activity experienced in a short time frame (whatever the time frame). The WCS corresponds to just a proportion of the total activity (Table 2) and classifying this as the worst or peak match demands underestimates the remaining physical requirements. The use of other

terms such as "peak locomotor demands" (Weaving et al., 2019), "duration-specific running demands" (Delaney, Duthie et al., 2016; Delaney, Thornton et al., 2016), or "high intensity phases", also used in the literature, is a better way to refer to this construct (although limitations such as univariate interpretations remain). While this may be semantic, avoiding poor terminology may limit over- and mis-interpretations.

There are also conceptual concerns on the use of WCS for training and setting targets. The idea that the WCS constitutes an optimal or sufficient physiological stimulus is a risky assumption, assuming that the optimal stimulus is the one provided by the game. Realistically, the optimal stimulus is one that enhances the targeted physiological of physical characteristics, yet there is no evidence that training programmes that induce the WCS have any effect on such characteristics. The use of an "average" positional or even individual WCS may prepare the players for an average demand, computationally cutting 50% of the WCS. Second, this also assumes that WCS has a role in crucial match moments and that being "psychologically" prepared may be advantageous. However, this is speculative. Nevertheless, once all the factors influencing these peak phases have been considered, these can be useful to understand the physical and physiological characteristics that can be targeted with specifically developed training programmes, together with the analysis of the whole match physical activities. But we argue that these analyses should not be used as benchmarks, that is to define the training quantity needed.

While we have discussed several limitations throughout, we also acknowledge that we only analysed a subset of the WCS metrics reported in the literature, however, these represent the most common (total distance) and those of most interest to practioners to target in training (high-speed running and sprinting).

Practical Applications

- Practitioners should refrain from referring to the WCS or peak demands when investigating univariate locomotor variables.
- Preparing athletes to meet a previous WCS or positional average may underprepare athletes for a true WCS.
- The true WCS should be recognized as a complex, composite construct that is actually an extreme internal response elicited via various combinations of physical and contextual factors.
- Practioners should be aware that the WCS does not occur concurrently for all metrics and players, which can inform their expectations of physical conditioning and representative training design.

Conclusion

Overall, the WCS when analysed as univariate metrics has poor consistency, particularly for metrics which are commonly targeted by practitioners in professional team sports i.e. high-intensity/threshold variables. As such, its use as benchmark for training is questionable. We propose that the WCS is better considered as a composite construct which may be induced by various combinations of physical activity and contextual factors, and which may be observable by appropriate measures of internal response. Finally, the concept of the WCS has not been previously well defined and the applicability to training and physiological adaptations as a result of training with WCS requires further investigation.

Disclosure statement

The authors report no conflict of interest.

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